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Survival analysis adjusting for measurement error in a cumulative total exposure variable: radon progeny to lung cancer mortality

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Summary

Measurement error in exposure variable in environmental epidemiology research is a very common occurrence and failure to properly adjust for such error typically biases exposure effect estimates towards the null. In this paper, we focus on risk set regression calibration method (RRC) which could correct for measurement error in time-varying exposure variables under Cox proportional hazards model and additive measurement error structure. We extend the RRC correction framework to cover both main/external and internal study design, both Cox proportional hazards and excess relative risk survival models, and both additive and multiplicative error structure. We further conduct simulation studies to investigate the finite-sample performance of the RRC correction method under extended scenarios of interest. We then apply the proposed methods in the New Mexico Uranium Miners study data with extended follow-up time until 2012 and report on estimated exposure effects without and with measurement error correction.

Key words: Exposure measurement error, Risk set regression calibration, Occupational radon progeny, Lung cancer mortality

1. Introduction

Radon (Rn) is a radioactive, odorless, colorless and tasteless gas and poses the second largest health risk to lung cancer in the US and first among non-smokers according to the Environmental Protection Agency's guide to the health risks of Radon. The relationship between exposure to radon (Rn) progeny and lung cancer development and mortality has been well documented in various epidemiologic and biological studies (Samet *and others* (1991); Tomášek *and others* (1999); Kusiak *and others* (1993); Howe *and others* (1987); Morgan and Samet (1986); Howe and Stager (1996); Heidenreich *and others* (2012); Lubin *and others* (1995); etc.). However, in these studies, investigators often need to estimate the levels of exposure to radon progeny using various sources over a long period of time, and thus the exposures are typically prone to both random and systematic errors (Pierce *and others* (1990); Stram

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and others (1999)). The existence of such exposure measurement error could bias the effect of exposure on outcome towards the null (Samet *and others* (1991); Spiegelman (2010); Spiegelman (2013); Stram *and others* (1999); Carroll *and others* (2006)).

Prentice (1982) first proposed the idea of regression calibration to address the issue of exposure measurement error in the setting of Cox proportional hazard (Cox-PH) survival model with mismeasured covariates. The method developed by Prentice (1982) is known as the ordinary regression calibration (ORC) method and the corresponding estimator is consistent in Cox-PH survival model only when certain conditions are satisfied, most importantly the rare disease assumption and the small effect assumption. Later, Xie *and others* (2001) introduced a new class of error correction method known as risk set regression calibration (RRC), which allows the measurement error model to change with time and loosens the rare disease assumption, although discussion is focused on time-invariant exposure. Liao *and others* (2011) and Liao *and others* (2018) continued the work on RRC method to extend to scenarios with measurement error in time-varying point exposures as well as functions of mis-measured exposure history, leveraging information from a main study/external validation study design, under the assumption of transportability of measurement error model between the main study and validation study population. This work also extended beyond classical additive Gaussian measurement error model but remains within additive error framework. Our work in this paper focuses on the RRC method and aims to further expand its scope to cover two different types of survival models, the Cox and excess relative risk (ERR) models, and two different types of measurement error model, the additive and multiplicative measurement error models.

In addition to log relative risk obtained from a Cox survival model, studies in the field of radiation epidemiology also commonly report ERR as a measure of exposure effect. Biologically-based mechanistic model exists to justifying the choice of ERR when assessing risk associated with Radon exposure (health Effects Of Exposure To Radon: BEIR VI. (1999)). A summary of empirical evaluation of 11 miners' studies further show that linear ERR model provide an adequate characterization of the relationship between Radon exposure and lung cancer, and there is no need to introduce non-linear ERR term except when cumulative exposure is extremely high. (Lubin *and others* (1994); Lubin *and others* (1995)). As a matter of fact, the recommended model when analyzing the health effects of exposure to Radon progeny is a form of piecewise linear ERR model (health Risks Of Radon and IV. (1988); health Effects Of Exposure To Radon: BEIR VI. (1999)). Most existing methods of adjusting for measurement error in exposure history were developed primarily for the Cox-PH survival model, including the RRC method so far and it is one of our goals to extend the RRC method further to be able to handle ERR models as well.

Another prominent feature in radiation epidemiology is that the distribution of observed radiation exposure is often skewed, which renders the Gaussian additive measurement error model problematic (Lubin *and others* (1995); Küchenhoff *and others* (2007); Bender *and others* (2005)). Prior work exploring the sources and structures of radon exposure measurement error has found that multiplicative measurement error structure empirically works well with real life data (Allodji *and others* (2012)) and this has the added benefit of maintaining an always-positive estimated true exposure range when correcting for measurement error. In addition, many literature reporting ERR as the exposure effect metric between radon progeny and cancer mortality adjusting for exposure measurement error works with the assumption that the measurement error is multiplicative and log-normal instead of additive and normal (Allodji *and others* (2012); Allodji *and others* (2012); Hoffmann *and others* (2017); Heidenreich *and others* (2012)). Similarly, most existing methods of adjusting for measurement error in exposure history, including and not limited to the RRC method have not been discussed in cases with non-additive measurement error structure and another one of our goals is to extend RRC into this scenarios as well.

Our methodology work is motivated by a real study population of interest. The New Mexico Uranium Miners study (Samet *and others* (1991)), which will be referred to as the Uminer study in this paper, included one of the largest number of underground Uranium miners in the US starting work from as early as 1957 and continuing until the mines closed in the 1980s. Initial results from the Uminer study analysis (Samet *and others* (1991)) were based on follow-up until 1985 with 68 lung cancer deaths in total and reported a linearly increasing risk of lung cancer mortality with increasing exposure to radon progeny. These results, while consistent with other studies of miners progeny to lung cancer, did not take

into account the likely existence of exposure measurement error. Additional study information from the Uminer population has always been made available with follow-up until 2012 and it is of great importance to perform the analysis with extended study time and adjustment for measurement error and report on the conclusion regarding radon progeny to lung cancer mortality. We have chosen cumulative total exposure as the exposure metric of interest and this is a function of potentially mis-measured exposure history.

In this paper, we have two primary goals. First, we aim to build upon prior risk set regression calibration methodology and expand it to accommodate a combination of scenarios with both Cox-PH and ERR survival models, as well as both additive and multiplicative measurement error models. Second, we intend to analyze the extended Uminer study cohort and report an updated effect estimation of exposure to occupational Radon progeny on lung cancer mortality.

Section 2 presents notation and model framework. Section 3 presents the method of risk set regression calibration and extension. Section 4 presents simulation studies of the extended method for a range of scenarios motivated by Uminers study. Section 5 shows the application and updated results of the extended method in the Uminers study, and Section 6 provides a discussion.

2. Notation and Framework

2.1 *Model Notation and Study Design*

Let i indicate the ith participant, t indicate age (or any consistent temporal metric), and $\boldsymbol{Z}(t)$ indicate a collection of fixed-at-baseline or time-varying covariates assumed to be free of measurement error. $\boldsymbol{Z}(t)$ could be considered as a set of potential risk factors or confounders. In the motivating Uminer study, we have information on 9 such covariates, including 1 smoking indicator variable (ever/never smoking), 2 race indicator variables (white/hispanic/other), 2 company indicator variables (3 regional companies in total) and 4 calendar period indicator variables (defined by each decade).

Let $c(t)$ indicate the true annual exposure, $C(t)$ indicate surrogate annual exposure. The unit of radon exposure in Uminer study is 100 WLM; 1 WLM is the cumulative exposure from breathing an atmosphere at a concentration of 1 working level (WL) for a working month of 170h, with 1 WL corresponding to 3,700 Bq, 1.3 $\times 10^8$ MeV or 20.8 $\times 10^{-5}$ J per cubic meter or air (*Tirmarche et al. 2010*). There are 5242 observations for 1163 participants between calendar year 1957-1967 without true annual exposure estimates, only surrogate annual exposure. Let $s(t)$ (`surf` variable) be an indicator of when the true annual exposure is unavailable and the surrogate annual exposure is available.

Our exposure measure of interest is a single cumulative total exposure $x(t)$ which is a function of annual exposure history $\{c(v), v \leqslant t\}$. Specifically, when we have all discrete exposure measurement time points, $x(t_k) = \sum_{j=1}^{k} c(t_j)I(t_j)$, where $I(t_j)$ is an indicator of whether the exposure variable is measured at age t_j, i.e. missing exposure data indicator. Instead of observing true annual exposure history, we sometimes only have access to surrogate annual exposure. As such, we further define observed annual exposure as $\tilde{C}(t_j) = c(t_j)(1-s(t_j)) + C(t_j)s(t_j)$ and defined observed cumulative total exposure variable $X(t_k) = \sum_{j=1}^{k} [c(t_j)(1-s(t_j)) + C(t_j)s(t_j)]I(t_j)$.

The motivating Uminer study is a main/internal validation study (MS/IVS) design. The main study consists of data $\{\{\tilde{C}_i(t_j), t_j \leqslant T_i\}, \{\boldsymbol{Z}_i(t_j), t_j \leqslant T_i\}, T_i, D_i\}, i = 1, \ldots, n_1$. $n_1 = 2337$ is the sample size of the main study, T_i is the follow up time defined as the minimum of potential failure time, potential censoring time (including death from other causes than the cause of interest) and administrative end of follow up T^*, D_i is an indicator of death due to lung cancer. The validation study consists of data $\{c_i(t_{ij}), C_i(t_{ij}), \boldsymbol{Z}_i(t_{ij}), T_i, D_i\}, i = 1, \ldots, n_2$. $n_2 = 863$ is the sample size of the internal validation study, which is contained within the main study population. From a preliminary explorations into the validation study data, we note that for any subject i included in the internal validation study, there could be more than 1 but no more than 5 time points when pairs of true and surrogate annual exposure variables are simultaneously available and included in the validation data, giving us $2,833$ pairs of true and surrogate

exposure values in total. The number of time points as well as the exact time points available vary across each subject. A special feature of the pairs of exposure data available in the Uminer validation data is that they all fall between the calendar year $1967-1976$ due to our definition of the true exposure variable and the development of exposure data collection procedure in the mining industry (more details in the study population section).

The risk set regression calibration method to correct for exposure measurement error was originally developed under the main/external validation study (MS/EVS) design and operates under the assumption of transportability between the measurement error model obtained from the validation study and the underlying measurement error structure in the main study. When dealing with the main/internal validation study design, we need a similar assumption (Carroll *and others* (2006)) that the internal validation population is representative of the main study population in the sense that the measurement error model estimated from the internal validation study will be indicative of the underlying measurement error structure in the entire main study population.

In addition, we also need the nondifferential measurement error, or surrogacy, assumption; i.e. given the true exposure history, the surrogate exposure history should have no association with the outcome, which means, for any general hazard function $\lambda(t)$, we have:

$$\lambda(t;\{c(v),v\leqslant t\},\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})=\lambda(t;\{c(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\}) \tag{2.1}$$

as well as the random censorship assumption (Prentice (1982)). The surrogacy assumption is typically guaranteed if the error-prone exposures are prospectively assessed, such as in a prospective cohort study.

Under the aforementioned set of assumptions, for any general hazard function $\lambda(t)$, we can bridge the gap between the true and surrogate exposure history like such:

$$\begin{aligned}&\lambda(t;\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})\\ =&\mathbb{E}_{\mathbf{c}}[\lambda(t;\{c(v),v\leqslant t\},\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})|T\geqslant t,\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\}]\\ =&\mathbb{E}_{\mathbf{c}}[\lambda(t;\{c(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})|T\geqslant t,\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\}]\end{aligned} \tag{2.2}$$

which could be written equivalently in terms of the exposure of interest, cumulative total exposure $x(t)$ as:

$$\begin{aligned}&\lambda(t;\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})\\ =&\mathbb{E}_{x(t)}[\lambda(t;x(t),X(t),\{\boldsymbol{Z}(v),v\leqslant t\})|T\geqslant t,\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\}]\\ =&\mathbb{E}_{x(t)}[\lambda(t;x(t),\{\boldsymbol{Z}(v),v\leqslant t\})|T\geqslant t,\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\}]\end{aligned} \tag{2.3}$$

2.2 *Survival Analysis Models*

In this paper, we are primarily interested in two common forms of survival models, the Cox proportional hazard model (Cox-PH) and the excess relative risk model (ERR) with linear term of exposure metric. It is possible to handle multiple exposure metrics in the model simultaneously, but for the purpose of our paper, we are interested in a single exposure metric, the cumulative total and thus only include a non-vector $x(t)$ in the model.

The Cox proportional hazards model holds the form

$$\lambda(t;\{c(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})=\lambda(t;x(t),\boldsymbol{Z}(t))=\lambda_0(t)exp(\boldsymbol{\beta_Z}Z(t))exp(\beta_x x(t))$$

where $\lambda_0(t)$ is the baseline hazard function and $\boldsymbol{\beta}=(\beta_x,\boldsymbol{\beta_Z})$ is a $(1+p)$ vector of coefficients, where $p=9$ in Uminer study. β_x estimates the increase in log relative risk for unit change of cumulative exposure and $\boldsymbol{\beta_Z}$ corresponds to the log risk ratio for one unit increase in each covariate.

The excess relative risk model with linear term of exposure metric holds the form

$$\lambda(t;\{c(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})=\lambda(t;x(t),\boldsymbol{Z}(t))=\lambda_0(t)exp(\boldsymbol{\beta_Z}Z(t))(1+\beta_x x(t))$$

where $\lambda_0(t)$ is the baseline hazard function and $\boldsymbol{\beta} = (\beta_x, \boldsymbol{\beta_Z})$ is a $(1+p)$ vector of coefficients, where $p = 9$ in Uminer study. β_x estimates the increase in excess relative risk for one unit increase in cumulative exposure and $\boldsymbol{\beta_Z}$ corresponds to the log risk ratio for one unit increase in each covariate.

Estimating effect coefficients under the Cox-PH model is a very common practice and various software are readily equipped with corresponding functions. However, fitting the ERR model poses some difficulties. Estimating effect coefficients under the ERR model is usually achieved in commercial statistical software, the most common being EPICURE, by researchers in the field of radiation epidemiology. Additional estimation methods have been proposed previously, including categorization of exposure and baseline covariates and performing contingency table analysis, which requires additional Poisson distribution assumption. These methods are relatively restrictive and generally requires additional assumptions about the risk models. To better compare with results obtained in the Cox-PH model, we decided to fit the ERR model mimicking the idea behind Cox-PH model and employ partial likelihood maximization. We give a brief description of the ERR model estimation procedure below.

Define risk set indicator in main study data as $Y_m(j, T_i) = I(T_j \geqslant T_i)$, which tells us whether subject j survives past subject i. Partial likelihood for the ERR model can be written as

$$L_i(\boldsymbol{\beta}) = \left\{ \frac{\lambda_i(T_i)}{\sum_{j=1}^{n_1} Y_m(j,t_i)\lambda_j(T_i)} \right\}^{D_i} = \left\{ \frac{\lambda_0(T_i) exp(\boldsymbol{\beta_Z} Z_i(T_i))(1+\beta_x x_i(T_i))}{\sum_{j=1}^{n_1} Y_m(j,t_i)\lambda_0(T_i) exp(\boldsymbol{\beta_Z} Z_j(T_i))(1+\beta_x x_j(T_i))} \right\}^{D_i}$$

$$= \left\{ \frac{exp(\boldsymbol{\beta_Z} Z_i(T_i))(1+\beta_x x_i(T_i))}{\sum_{j=1}^{n_1} Y_m(j,t_i) exp(\boldsymbol{\beta_Z} Z_j(T_i))(1+\beta_x x_j(T_i))} \right\}^{D_i}$$

$$\Rightarrow l_i(\boldsymbol{\beta}) = log L_i(\boldsymbol{\beta}) = D_i[\boldsymbol{\beta_Z} Z_i(T_i) + log(1+\beta_x x_i(T_i)) - log(\sum_{j=1}^{n_1} Y_m(j,t_i) exp(\boldsymbol{\beta_Z} Z_j(T_i))(1+\beta_x x_j(T_i)))]$$

$$\Rightarrow l(\boldsymbol{\beta}) = \sum_{i=1}^{n_1} D_i[\boldsymbol{\beta_Z} Z_i(T_i) + log(1+\beta_x x_i(T_i)) - log(\sum_{j=1}^{n_1} Y_m(j,t_i) exp(\boldsymbol{\beta_Z} Z_j(T_i))(1+\beta_x x_j(T_i)))]$$

$$\Rightarrow U(\boldsymbol{\beta}) \triangleq \frac{\partial l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}} = \left(\frac{\partial l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta_x}}, \frac{\partial l(\boldsymbol{\beta})}{\partial \boldsymbol{\beta_Z}} \right)^T = \mathbf{0}$$

The procedure of fitting the ERR survival model is now transformed into the problem of finding $\hat{\boldsymbol{\beta}}$ such that it maximizes $l(\boldsymbol{\beta})$ or equivalently such that $U(\hat{\boldsymbol{\beta}}) = \mathbf{0}$ holds. This is now a maximization computation problem and could employ various readily available functions in statistical softwares. We have written code in R statistical software that handles ERR model fitting and it could be provided upon request.

In survival analysis, the time points of most interest are unique survival times of cases, i.e. T_i for subjects with $D_i = 1$. In the Uminer study, we observe 132 deaths due to lung cancer with 39 unique death ages and there are no more than 5 deaths tied at each unique failure ages. Since the number of ties is relatively small compared to the size of most risk sets, it is reasonable to use the Breslow correction for ties in failure times in the partial likelihood maximization process for the purpose of this paper.

2.3 *Measurement Error Models*

In this paper, we shall look at two common types of underlying true exposure measurement error models, additive linear and multiplicative linear.

We write the general form of an additive linear measurement error model for $c(t)|C(t), \boldsymbol{Z}(t)$ as

$$\mathbb{E}[c(t_k)|C(t_k), \boldsymbol{Z}(t_k), T \geqslant t_k] = \alpha_0(t_k) + \alpha_1(t_k)C(t_k) + \boldsymbol{\alpha_2}(t_k)\boldsymbol{Z}(t_k) \tag{2.4}$$

and a multiplicative linear measurement error model for $c(t)|C(t), \boldsymbol{Z}(t)$ as

$$log(c(t_k))|C(t_k), \boldsymbol{Z}(t_k), T \geqslant t_k = \alpha_0(t_k) + \alpha_1(t_k)log(C(t_k)) + \boldsymbol{\alpha_2}(t_k)log(\boldsymbol{Z}(t_k)) + e_M \tag{2.5}$$

where $e_M \sim \mathcal{N}\big(0,\ \sigma^2_M(t_k)\big)$, and $\boldsymbol{\alpha}_k = (\alpha_0(t_k), \alpha_1(t_k), \boldsymbol{\alpha_2}(t_k))$ and σ^2_M are estimated in the validation study data.

If we remove the conditional term $T \geqslant t_k$ from the measurement error models, we are equivalently making the assumption that the measurement error model remains the same across all risk sets defined by unique failure times and we could then estimate a single set of $\boldsymbol{\alpha} = (\alpha_0, \alpha_1, \boldsymbol{\alpha}_2)$ in the entire validation study data. Since we are unwilling to make such a strong assumption, we will correspondingly obtain a different set of $\boldsymbol{\alpha}_k = (\alpha_0(t_k), \alpha_1(t_k), \boldsymbol{\alpha_2}(t_k))$ in each risk set of the validation study data defined by unique failure times in the main study.

Since

$$\begin{aligned}
\log\big[E\big(c(t_k) \mid C(t_k), Z(t_k)\big)\big] &= \log\Big(E\big[\exp(\log c(t_k)) \,\big|\, C(t_k), Z(t_k)\big]\Big) \\
&= E\big[\log(c(t_k)) \mid C(t_k), Z(t_k)\big] + \frac{1}{2}\, Var\big(\log(c(t_k)) \mid C(t_k), Z(t_k)\big) \\
&= \alpha_0(t_k) + \alpha_1(t_k)\log C(t_k) + \boldsymbol{\alpha}_2(t_k)\log Z(t_k) + \frac{\sigma^2_M(t_k)}{2},
\end{aligned}$$

$$E\big(c(t_k) \mid C(t_k), Z(t_k)\big) = \exp\left\{\alpha_0(t_k) + \frac{\sigma^2_M(t_k)}{2}\right\} C(t_k)^{\alpha_1(t_k)}\, Z(t_k)^{\boldsymbol{\alpha}_2(t_k)}.$$

3. Risk Set Regression Calibration Methodology

3.1 *Extending Risk Set Regression Calibration Method*

The classic risk set regression calibration (RRC) method can re-calibrate the measurement error model within each risk set in the validation study data and was previously developed to handle modeling the Cox survival model of time-varying functions of exposure history with additive measurement error in a main/external validation study(MS/EVS) study design. (Liao *and others* (2011); Liao *and others* (2018))

We will discuss key steps of RRC method and focus on what changes when we extend to cover ERR survival model, multiplicative measurement error model and main/internal validation study scenario.

As previously discussed, under IVS representativeness, nondifferental measurement error and random censorship assumptions, we have:

$$\begin{aligned}
&\lambda(t; \{\tilde{C}(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}) \\
&= \mathbb{E}_c[\lambda(t; \{\tilde{C}(v), v \leqslant t\}, \{c(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}) | T \geqslant t, \{\tilde{C}(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}] \\
&= \mathbb{E}_c[\lambda(t; \{c(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}) | T \geqslant t, \{\tilde{C}(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}]
\end{aligned} \tag{3.6}$$

For Cox-PH survival model, we could further write:

$$\begin{aligned}
&\lambda(t; \{C(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}) \\
=&\lambda_0(t) exp(\beta_{\mathbf{Z}} Z(t)) \mathbb{E}_{x(t)}[exp(\beta_x x(t)) | T \geqslant t, \{C(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}] \\
\approx&\lambda_0(t) exp(\beta_{\mathbf{Z}} Z(t)) exp(\beta_x \mathbb{E}_{x(t)}[x(t) | T \geqslant t, \{C(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}])
\end{aligned} \tag{3.7}$$

The approximation in the final step holds exactly when the distribution of $(x(t)|\{C(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\})$ is normal with homoscedastic covariance matrix Σ_x in each risk set. It could also hold approximately if (1) the distribution of $(x(t)|\{C(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\})$ is normal with small $\beta_x^T \Sigma_x(C(t), \mathbf{Z}(t))\beta_x$; or (2) β_x is small with either homoscedastic covariance matrix Σ_x or small $\beta_x^T \Sigma_x(C(t), \mathbf{Z}(t))\beta_x$ in each risk set, where $\Sigma_x(C(t, \mathbf{Z}(t))) = Var(x|C, \mathbf{Z})$ (Xie *and others* (2001)). We will loosely refer to this set of

assumptions as the 'small effect' assumption, which is required when we apply the regression calibration method to Cox-PH survival model.

For ERR survival model, we could similarly write:

$$\begin{aligned}&\lambda(t;\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\})\\=&\lambda_0(t)exp(\beta_{\mathbf{Z}}Z(t))\mathbb{E}_{x(t)}[(1+\beta_x x(t))|T\geqslant t,\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\}]\\=&\lambda_0(t)exp(\beta_{\mathbf{Z}}Z(t))(1+\beta_x\mathbb{E}_{x(t)}[x(t)|T\geqslant t,\{C(v),v\leqslant t\},\{\boldsymbol{Z}(v),v\leqslant t\}])\end{aligned}\tag{3.8}$$

Unlike for the Cox-PH survival model, the final step always holds exactly with no additional assumption needed.

Regardless of the measurement error model form, additive or multiplicative, we perform GEE estimation in each risk set of the internal validation study to obtain the error parameters $\hat{\boldsymbol{\alpha}}(t_k)=(\hat{\alpha_0}(t_k),\hat{\alpha_1}(t_k),\hat{\boldsymbol{\alpha}}_{\mathbf{2}}(t_k))$ defined by unique event times t_k and obtain $\hat{\boldsymbol{\alpha}}=(\hat{\boldsymbol{\alpha}}(t_{k_1}),\ldots,\hat{\boldsymbol{\alpha}}(t_{k_R}))$. Plugging in estimated error parameters into respective measurement error models, we obtain estimated true annual exposure variable $\hat{c}(t_k)$ and further obtain estimated true cumulative total exposure variable $\hat{x}(t_k)=\sum_{j=1}^{k}[c(t_j)(1-s(t_j))+\hat{c}(t_j)s(t_j)]I_{t_j}$. We then further fit the respective Cox-PH or ERR survival model with $\hat{x}(t_k)$ using partial likelihood maximization process and obtain the target exposure effect coefficient $\hat{\beta}_{x,RRC}$ within $\hat{\boldsymbol{\beta}}_{RRC}$.

In its essence, for Cox-PH survival model $\hat{\beta}_{x,RRC}$ solves the following score estimating equation:

$$\sum_{i=1}^{n_1}\int_0^{T^*}\left\{\begin{pmatrix}\hat{x}_i(\hat{\boldsymbol{\alpha}},t)\\\mathbf{Z}_i(t)\end{pmatrix}-\frac{\mathbf{S}^{(1)}_{Cox}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)}{S^{(0)}_{Cox}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)}\right\}N_i(dt)=0\tag{3.9}$$

where

$$S^{(0)}_{Cox}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)=\frac{1}{n_1}\sum_{i=1}^{n_1}Y_m(i,t)exp(\boldsymbol{\beta}_{\boldsymbol{Z}}\mathbf{Z}(t))exp(\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t))\tag{3.10}$$

and

$$S^{(1)}_{Cox}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)=\frac{1}{n_1}\sum_{i=1}^{n_1}Y_m(i,t)\begin{pmatrix}\hat{x}_i(\hat{\boldsymbol{\alpha}},t)\\\mathbf{Z}_i(t)\end{pmatrix}exp(\boldsymbol{\beta}_{\boldsymbol{Z}}\mathbf{Z}(t))exp(\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t))\tag{3.11}$$

Similarly, for ERR survival model $\hat{\beta}_{x,RRC}$ will solve the following:

$$\sum_{i=1}^{n_1}\int_0^{T^*}\left\{\begin{pmatrix}\frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}\\\mathbf{Z}_i(t)\end{pmatrix}-\frac{\mathbf{S}^{(1)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)}{S^{(0)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)}\right\}N_i(dt)=0\tag{3.12}$$

where

$$S^{(0)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)=\frac{1}{n_1}\sum_{i=1}^{n_1}Y_m(i,t)exp(\boldsymbol{\beta}_{\boldsymbol{Z}}\mathbf{Z}(t))(1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t))\tag{3.13}$$

and

$$S^{(1)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)=\frac{1}{n_1}\sum_{i=1}^{n_1}Y_m(i,t)\begin{pmatrix}\frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}\\\mathbf{Z}_i(t)\end{pmatrix}exp(\boldsymbol{\beta}_{\boldsymbol{Z}}\mathbf{Z}(t))(1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t))\tag{3.14}$$

where counting process $N_i(t)=I(T_i\leqslant t,D_i=1)$, $N_i(dt)=1$ if subject i fails at time t and $N_i(dt)=0$ otherwise, $Y_m(i,t)=I(T_i\geqslant t_{k_r})$ is the risk process indicator in the main study data with $r=1,\ldots,R$ and R being the number of unique failure times in the main study data.

Asymptotic consistency and normality discussions of the estimator $\hat{\boldsymbol{\beta}}_{RRC}$ are similar to the works of Liao *and others* (2011) and presented in the supplementary materials. The variance of $\hat{\boldsymbol{\beta}}_{RRC}$ is obtained via a sandwich form

$$\hat{Var}[\hat{\boldsymbol{\beta}}_{RRC}]=\frac{1}{n}\hat{I}_{\beta}^{-1}\hat{H}_{\beta,\alpha}\hat{I}_{\beta}^{-1}$$

where the exact formula is also included in the supplementary materials.

3.2 *Procedures of Risk Set Regression Calibration Method*

Step 1 Order all R unique event times in the main study data as $t_{k_1} < t_{k_1} < \cdots < t_{k_R}$. In the event of tied cases at each event times, employ necessary methods to differentiate between single and repetitive event times. For our purposes, the Breslow method is sufficient.

Step 2 Generate risk process indicator in the main study data as $Y_m(i, t_{k_r}) = I(T_i \geqslant t_{k_r})$, where $r = 1, \ldots, R$, $i = 1, \ldots, n_1$. Define R risk sets as the subjects still at risk of the outcome at time t_{k_r} in the main study population $R_{m(t_{k_r})}$.

Step 3 Generate risk process indicator in the validation study data as $Y_v(i, t_{k_r}) = I(T_i \geqslant t_{k_r})$, where $r = 1, \ldots, R$, $i = 1, \ldots, n_2$. Define R risk sets as the subjects still at risk of the outcome at time t_{k_r} in the validation study population $R_{v(t_{k_r})}$.

Step 4 In each validation data risk set $R_{v(t_{k_r})}$, fit a chosen form of measurement error model for $c(t_{k_r})|C(t_{k_r}), \mathbb{Z}(t_{k_r}), T \geqslant t_{k_r}$, either additive or multiplicative and obtain estimated parameters $(\hat{\alpha_0}(t_{k_r}), \hat{\alpha_1}(t_{k_r}), \hat{\boldsymbol{\alpha_2}}(t_{k_r}))$, as well as $\hat{\sigma}^2_M$ in the case of multiplicative error model.
Note: given the usually small sample size in validation data, towards the later event times, defined risk sets generally won't contain enough data entries to estimate measurement error model successfully and consistently. In these cases, one solution is repeatedly combine any risk set containing fewer than a pre-determined number of entries with the nearest risk set until each final risk set contains enough information for stable measurement error estimation. This adjustment allows for repeated exposure measurements from the same subject to fall under the same validation study risk set and to counter additional correlation introduced by repeated measurements, we employ GEE estimation instead of simple linear regression to obtain $(\hat{\alpha_0}(t_{k_r}), \hat{\alpha_1}(t_{k_r}), \hat{\boldsymbol{\alpha_2}}(t_{k_r}))$ and correspondingly use empirical sandwich variance estimators as well, which produces consistent coefficient estimators even under misspecification of covariance matrix.

Step 5 Plug in estimated parameters $(\hat{\alpha_0}(t_{k_r}), \hat{\alpha_1}(t_{k_r}), \hat{\boldsymbol{\alpha_2}}(t_{k_r}))$, and $\hat{\sigma}^2_M$ in the case of multiplicative error model, into respective measurement error model forms and obtain $\hat{x}_i(t_{k_r})$ for each subject i in each risk set $R_{m(t_{k_r})}$ in the main study, where $r = 1, \ldots, R$, $i = 1, \ldots, n_1$.

Step 6 Fit respective Cox-PH or ERR survival model on $(\hat{x}_i(t_{k_t}), \mathbf{Z}_i(t_{k_r}), Y_m(i, t_{k_r}), T_i, D_i)$ and obtain $\hat{\beta}_{x,RRC}$. The model estimation procedure utilizes partial maximum likelihood estimation procedure.

4. Simulation Study

4.1 *Simulation 1: Validity of RRC Method Extended to Cox-PH/ERR Survival Models Under Additive/Multiplicative Measurement Error Models*

We perform finite sample simulation analysis to showcase the effect of the RRC procedure detailed in Section 3.2 on the correction of measurement error in the extended scenarios of both Cox-PH and ERR survival models and both additive and multiplicative linear measurement error models.

We generate surrogate annual exposure measurements $\boldsymbol{C}_{n_1 \times T} \sim MVN(\boldsymbol{\mu_C}, \boldsymbol{\Sigma_C})$ if assuming additive measurement error and $\boldsymbol{C}_{n_1 \times T} \sim log - MVN(\boldsymbol{\mu_C}, \boldsymbol{\Sigma_C})$ if assuming multiplicative measurement error for $n_1 = 2,400$ subjects and $T^* = 55$ years and generate true annual exposure measurements $c_{ij} = \gamma_0 + \gamma_1 C_{ij} + \epsilon$ if additive and $log(c_{ij}) = \gamma_0 + \gamma_1 log(C_{ij}) + \epsilon$ if multiplicative where $i = 1 \ldots, n_1$ indicating subject and $j = 1, \ldots, T^*$ indicating time metric. The values of $\boldsymbol{\mu_C}, \boldsymbol{\Sigma_C}, \gamma_0, \gamma_1, cor(c, C)$ are chosen to approximate the estimated corresponding parameters in Uminer study data. For the purposes of simulation study, we do not include any other baseline covariates that may be risk factors or potential confounders to the event.

It is important to note that annual exposure measurements, whether surrogate or true, from the same subject across different calendar years are likely to be correlated. We mimic this by choosing different structures of covariance matrix $\boldsymbol{\Sigma_C}$ when generating surrogate annual exposure, ranging from independent covariance structure, AR-1 covariance structure, damped exponential covariance structure

and finally, sampling with replacement from the actual Uminer exposure data. All parameters involved in the specification of the covariance matrix are again chosen to approximate the estimated corresponding parameters in Uminer study data.

We then choose comparable pairs of Cox-PH and ERR effect coefficients (see explanation in supplementary materials) β_{Cox} and β_{ERR} and simulate survival times T_i from corresponding models in the following steps. The steps below is written out for ERR model and replacing $(1+\beta x_{ik})$ with $exp(\beta x_{ik})$ shall give us the necessary formulas for Cox-PH model (Liao *and others* (2011, 2018)).

Steps of generating survival data with time-varying cumulative total exposure under the ERR surviva model:

Step 1 Calculate time-varying cumulative total exposure, both true $x_{ij} = \sum_{k=1}^{j} c_{ik}$ and surrogate $X_{ij} = \sum_{k=1}^{j} C_{ik}$

Step 2 Calculate cumulative incidence function, defining $x_{i0} = 0$ and chooseing Weibull(ν, θ) to be the base hazard function

$$F_{ij} = 1 - exp\{-\nu^{\theta}(\sum_{k=0}^{j-1}(1+\beta x_{ik})[(k+1)^{\theta} - k^{\theta}])\}$$

(ν, θ) are chosen such that the cumulative incidence rate in the generated simulation data reaches a pre-determined level, in our case approximately $5-10\%$.

Step 3 Generate $U_i \sim U(0,1)$

Step 4 Find the time period j such that $F_{ij} \leqslant U_i < F_{i,j+1}$ and calculate

$$T_i^0 = \left\{ j^{\theta} - 1/(1+\beta x_{ij}) \left(\nu^{-\theta} log(1-U_i) + \sum_{k=0}^{j-1}(1+\beta x_{ik})[(k+1)^{\theta} - k^{\theta}] \right) \right\}^{1/\theta}$$

If $U_i > F_{i,t^*}$, then we set $T_i^0 = 2T^*$

Step 5 Generate censoring time $V \sim exp(\lambda)$, for this set of simulations we assume no censoring effect. Calculate failure time $T_i = min(T_i^0, V_i, T^*)$

Step 6 Create event indicator variable $D_{ij} = I(T_i <= j)$ and drop all measurements for subject i after $D_{ij} = 1$ for the first time

At this point, the simulated main study data consists of $subject = i, time = j, c_{ij}, C_{ij}, x_{ij}, X_{ij}, T_i, D_{ij}$. Now we randomly sample $n_2 = 800$ into a validation study and again randonly select only $3,000$ pairs of (c_{ij}, C_{ij}) measurement entries with no more than 8 entries from the same subject, which is usually met without manual adjustment.

We now obtain error-prone or naive estimates of exposure effect $\hat{\beta}_{naive}$ by fitting the corresponding Cox/ERR model of $D_{ij} \sim C_{ij}$ and apply RRC method to calculate adjusted $\hat{C_{ij}}$ and obtain RRC-corrected estimates of exposure effect $\hat{\beta}_{RRC}$ by fitting $D_{ij} \sim \hat{C_{ij}}$. Repeat the process for $B = 2,000$ times. We present the relative bias, estimated standard error, ratio of standard error to theoretical standard deviation and empirical 95% coverage rate of the naive and RRC-corrected estimated exposure effect in the Table 1.

General observation from Table 1 is that the application of RRC correction for measurement error improves the relative bias and coverage of the estimated exposure effect significantly, although in finite sample study such as this, the corrected effect is still slightly biased towards the null. Relative bias percentage is much lower after RRC correction and coverage rate is very near the 95% theoretical mark. This correction method works well for both ERR and Cox survival models and for both additive and multiplicative measurement error. In addition, as the correlation structure gets more complicated between the annual measurements from the same subject, both the naive and RRC-corrected methods become

Table 1: Summary Results of Simulation Study of Naive VS RRC-Corrected Effect Estimates of Cumulative Total Exposure

Generation of C	**Relative bias (%)**		**Estimated SE $\hat{se}(\hat{\beta})$**		**SE/SD**		**95% CI coverage (%)**	
	Naive	RRC	Naive	RRC	Naive	RRC	Naive	RRC
Cox+Additive	Cox true $\beta = log(1.5) \approx 0.405$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-46.97	-3.32	0.054	0.093	0.997	0.994	0.00	94.80
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-52.40	-4.55	0.056	0.105	0.996	0.992	0.00	94.30
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-52.15	-4.80	0.055	0.105	0.996	0.992	0.00	94.25
Resample from data	-53.39	-6.77	0.062	0.119	0.994	0.992	0.00	93.85
Cox+Multiplicative	Cox true $\beta = log(1.5) \approx 0.405$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-48.22	-3.64	0.059	0.096	0.996	0.995	0.00	94.55
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-52.69	-4.90	0.059	0.111	0.996	0.993	0.00	94.25
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-52.74	-5.04	0.061	0.112	0.995	0.992	0.00	94.15
Resample from data	-53.88	-7.12	0.065	0.121	0.994	0.992	0.00	93.70
Cox and ERR true β are chosen to produce approximately comparable risk model								
ERR+Additive	ERR true $\beta \approx 0.423$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-49.34	-4.78	0.077	0.106	0.997	0.994	0.00	94.40
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-53.41	-6.14	0.086	0.111	0.997	0.994	0.00	94.25
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-53.98	-6.99	0.089	0.110	0.997	0.994	0.00	94.10
Resample from data	-56.50	-9.26	0.112	0.119	0.995	0.991	0.00	93.60
ERR+Multiplicative	ERR true $\beta \approx 0.423$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-50.12	-4.96	0.083	0.110	0.997	0.992	0.00	94.20
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-55.56	-6.86	0.092	0.114	0.996	0.991	0.00	94.05
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-56.03	-7.80	0.091	0.113	0.995	0.991	0.00	94.00
Resample from data	-58.87	-9.93	0.119	0.124	0.994	0.989	0.00	93.55

†Cox-PH survival model $\lambda(t) = \lambda_0(t)exp(\beta x(t))$ with true $\beta_{Cox} = log(1.5) \approx 0.405$ and comparable ERR survival model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{ERR} \approx 0.423$

‡Main study sample size n_1=2400, Validation study sample size n_2=800, Study duration T=55, Simulation repetition number B=2000, CIR between $5 - 10\%$

more biased. In realistic epidemiological studies, it is unreasonable to assume independent annual measurements as a subject is likely to remain in the same occupational environment for a consistent number of years, thus introducing correlation to their exposures from year to year. This will undoubtedly affect the RRC method's ability to completely correct for error in a long history of measurements but it is a valid and useful method and should work better with increasing sample size.

Results presented in Table 1 is when true underlying exposure effect is relatively small $\beta_{Cox} \approx 0.405$, satisfying the loose 'small effect' assumption needed to successfully apply RRC correction method to Cox-PH survival model. We also looked at a collection of finite-sample simulation results for when the loose 'small effect' assumption is broken, namely when $\beta_{Cox} \approx 1.609$ and comparable $\beta_{ERR} \approx 1.905$. Additional simulation results are included in supplementary materials.

4.2 *Simulation 2: Validity of RRC Method Under Model Misspecification*

In previous simulations, we performed RRC correction under the assumption that we are aware of the true underlying effect and measurement error models form and therefore fit the correspondingly correct models in the procedure. We are also interested in exploring the behavior of RRC correction method

in the presence of model misspecification. In this set of simulations, we are no longer concerned with mimicking features of the real Uminer study data.

Different from Simulation 1, we start with generating the true annual exposure $\boldsymbol{c}_{n_1,T} \sim MVN(\boldsymbol{\mu_c}, \boldsymbol{\Sigma_c})$ for $n_1 = 2,000$ subjects and $T^* = 60$ years, where $\boldsymbol{\Sigma_c}$ is a compound symmetry covariance structure with $\sigma^2 = 15^2$ and $\rho = 0.5$. We then generate surrogate yearly exposure level $c_{ij} = \gamma_0 + \gamma_1 C_{ij} + \epsilon$ if additive and $log(c_{ij}) = \gamma_0 + \gamma_1 log(C_{ij}) + \epsilon$ if multiplicative where $i = 1, \ldots, n_1$ and $j = 1, \ldots, T^*$. γ_0, γ_1 are chosen to achieve the desired $\boldsymbol{\mu_C}, \boldsymbol{\Sigma_C}$ and pre-set $\rho = corr(c, C)$ ranging from 0.3, 0.6, 0.9, representing a decreasing amount of measurement error.

We then choose comparable pairs of Cox-PH and ERR effect coefficients (again see explanation in supplementary materials) β_{Cox} and β_{ERR} and simulate survival times T_i from the same process described in Simulation 1 and obtain a main study data set consisting of $subject = i, time = j, c_{ij}, C_{ij}, x_{ij}, X_{ij}, T_i, D_{ij}$. Now we randomly sample $n_2 = 500$ subjects into a validation study data set and again, keep no more than 8 entries from the same subject.

Pretending we have no credible knowledge of the true underlying generation model, we perform the following model fitting: (1) a naive Cox model of $D_{ij} \sim C_{ij}$; (2) a naive ERR model of $D_{ij} \sim C_{ij}$; (3) a RRC-corrected Cox model of $D_{ij} \sim \hat{C_{ij}}$, where $\hat{C_{ij}}$ is obtained assuming additive measurement error; (4) a RRC-corrected Cox model of $D_{ij} \sim \hat{C_{ij}}$, where $\hat{C_{ij}}$ is obtained assuming multiplicative measurement error; (5) a RRC-corrected ERR model of $D_{ij} \sim \hat{C_{ij}}$, where $\hat{C_{ij}}$ is obtained assuming additive measurement error; (6) a RRC-corrected ERR model of $D_{ij} \sim \hat{C_{ij}}$, where $\hat{C_{ij}}$ is obtained assuming multiplicative measurement error. Repeat the process for $B = 2,000$ times. We present the relative bias from all 6 estimated exposure effects in Table 2.

For each scenario, as $\rho(c, C)$ increases, the degree of measurement error decreases and the performance of RRC increases accordingly. Focusing on the bold cells where both risk and error model is correctly specified, RRC corrected results in those cases generally produce the best results. Even when either risk or error model is mis-specified, applying the RRC correction method still produces significantly better results than the naive estimates. An additional observation is that when Cox model is mis-specified as the ERR model, the RRC has a tendency to overcorrect for the error, leading to bias away from the null rather than towards the null; conversely, when ERR model is mis-specified as the Cox model, the RRC is less powerful in correcting for the error with a significant remaining amount of bias in the corrected estimates. It would seem that misspecification of the risk model results is more detrimental to the effect of the RRC method compared to the misspecification of the error model.

5. Data Example

5.1 *Uminer Study Population*

In 1977 the University of New Mexico initiated a series of investigations to examine morbidity and mortality among Uranium miners in New Mexico's principal mining area, and 4,044 men working in underground Uranium mines in New Mexico on or after 1957 were originally screened for the study. Of those, 398 men were excluded because they had less than a year mining experience in New Mexico, 172 were excluded with missing birth year, exposure history or followup information and 5 were excluded because they were of an race other than white or native American, resulting in a total of 3,469 men analyzed in previous studies by Samet *and others* (1986), Samet *and others* (1989) and Samet *and others* (1991).

Since then, followup on this cohort with respect to lung cancer mortality continued on. To ascertain vital status, the records of the New Mexico Uranium mining companies were combined with those from the New Mexico Tumor Registry, the New Mexico Bureau of Vital Records and Health Statistics, and the National Death Index. Vital status was determined through February 28, 2012, providing over 133,659 person-years during which 251 deaths from lung cancer occurred. Cause of death was coded using the International Classification of Diseases (ICD), and lung cancer mortality was defined by the ICD code for malignant neoplasms of the trachea, bronchus, and lung (ICD-8: 162; ICD-9: 162; ICD-10: C33-C34).

Table 2: Relative Bias Results of Exposure Effect Estimates of Cumulative Total Exposure With Model Misspecification (**Bold** cells with correct model specification)

$\rho(\mathbf{c}, \mathbf{C})$	**Relative bias (%)**					
	Naive		RRC Corrected			
	Cox	ERR	Cox.A	Cox.M	ERR.A	ERR.M
Cox+Additive	Cox true $\beta = 1$, additive independent error					
0.3	-33.5	-28.3	**-4.4**	-5.7	5.9	6.6
0.6	-32.8	-27.9	**-4.0**	-5.5	5.6	6.1
0.9	-31.4	-27.2	**-3.8**	-5.4	5.3	5.7
Cox+Multiplicative	Cox true $\beta = 1$, multiplicative independent error					
0.3	-34.7	-29.0	-6.1	**-5.7**	4.8	4.3
0.6	-33.9	-28.4	-5.6	**-5.3**	4.5	4.1
0.9	-32.6	-27.8	-5.4	**-5.0**	4.0	3.7
Cox and ERR true β are chosen to produce approximately comparable risk model						
ERR+Additive	ERR true $\beta = 1.109$, additive independent error					
0.3	-42.1	-35.8	-14.9	-15.2	**-5.5**	-6.6
0.6	-40.6	-33.0	-14.5	-14.9	**-5.1**	-6.4
0.9	-38.9	-31.2	-14.2	-14.7	**-4.6**	-6.2
ERR+Multiplicative	ERR true $\beta = 1.109$, multiplicative independent error					
0.3	-43.0	-36.2	-16.8	-16.0	-6.7	**-6.3**
0.6	-41.7	-33.4	-15.8	-15.6	-5.8	**-6.1**
0.9	-40.4	-31.9	-14.7	-15.2	-4.8	**-5.7**

[§]Cox-PH survival model $\lambda(t) = \lambda_0(t)exp(\beta x(t))$ with true $\beta_{Cox} = 1$ and comparable ERR survival model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{ERR} \approx 1.109$

[¶]Main study sample size n_1=2000, Validation study sample size n_2=500, Study duration T=60, Simulation repetition number B=2000, CIR about 10%

In this study, exposure to radon progeny during underground mining was assessed differently depending on the calendar time period. Prior to 1957, exposure estimates were based on a limited number of measurements assigned to large units, such as the mine or the state, sometimes averaged over several calendar years. Between 1957 and 1967, environmental area sampling data were available for the NM mines at the level of mine section within a company, which we have denoted in this paper as error-prone or surrogate yearly exposure data C. Further details on this portion of the exposure assessment were reported in previous literature (Samet *and others* (1991)). After 1967, mining companies provided individual annual exposure estimates based on a more dense sampling strategy within the mines, which we have denoted in this paper as true yearly exposure data c. These are regarded as the gold standard because these measurements are considered the most accurate and are preferred over all other available methods of exposure intensity estimation. Unit for all exposure measurements is 100 WLM.

For the main study population in our measurement error correction analysis, we included the 2,337 of 3,469 miners whose exposure data consisted solely of company/section samples C and/or individual annual exposure estimates c. In this group, 132 deaths from lung cancer occurred during 88,405 person-years of follow-up between 1957 and 2012. Only 7 lung cancer deaths were in never smokers.

For the validation study population in our measurement error correction analysis, we require data in which individual annual exposure measurements and exposure estimates from surrogate sources are available for the same miner in the same year. The validation study consists of 2,833 pairs of annual company/section samples and individual annual exposure measurements taken between 1967 and 1976 among 862 miners.

We provide a summary of baseline characteristics of the main study population of 2,337 miners and validation study population of 862 miners used in our analysis in Table 3. Measured participant characteristics include age, calendar year, ethnicity (Hispanic, non-Hispanic white, and American Indian) and 3 companies, which were the largest operating in the mining district. It should be noted that the distributions of baseline characteristics included in the study are reasonably similar in main and validation study populations, which is a good indication that the validation study population is representative of the main study population.

Table 3: Summary of Baseline Characteristics in Uminer Main and Validation Study Population

	Main Study	**Validation Study**
Years of exposure data	1957-2012	1967-1976
Participant/Person-year	2,337/88,405	862/2,833
Company-section measurements (C): Mean(SD)[1]	27.1(25.4)	4.5(3.8)
Participant/Person-year	1,163/5,242	862/2,833
Individual estimates (c): Mean(SD)[1]	1.7(1.9)	1.9(2.2)
Participant/Person-year	1,858/12,598	862/2,833
Not working (c**=0**): Participant/Person-year	2,287/71,003	0/0
Ethnic distribution: (White, Hispanic, Native-American)%	(52,43,5)	(40,51,9)
Birth year: (<1925, 1925-1934, 1935-1944, >1944)%	(13,21,30,37)	(13,23,32,32)
Cigarette smoking: (Never, Ever)%	(26,74)	(25,75)
Company: (5,6,12)%	(24,67,9)	(15,73,12)
Age at entry into cohort: Mean(SD)	28.0(8.2)	28.3(8.0)
(<20,20-30,30-40,⩾40)%	(10,58,23,10)	(6,60,24,11)

[‖]Unit for exposure measurement is 100 WLM.

5.2 *Effects of Exposure to Radon Progeny on Lung Cancer Mortality*

We fit a variety of different models and produce the following results in Table 4: naive VS RRC-corrected, Cox VS ERR, additive VS multiplicative and adjusting for no other covariates in the data except age VS adjusting for all other covariates in the data including age.

In the Uminer data population, we adjust for measurement error using RRC procedure detailed in Section 3.2. There are 132 lung cancer death and 39 unique death ages in the main study population, with no more than 5 ties at each death ages. When creating risk sets in the validation study data, there are 0 data entries at and after the 22nd death age. Additionally, risk sets from the 13th to the 21st death age contain fewer than 20 data entries. Starting from the latest death age, we work backwards and combine risk sets with the previous one until all risk sets contain more than 20 data entries. After combination, we end up with 14 risk sets in total.

When we analyze the multivariate model including all baseline covariates available in the Uminer study data, we include all of them simultaneously in the survival and measurement error models. This avoids the subject matter discussion of which ones can directly impact exposure level, lung cancer death or the measurement error structure. Including all potential risk factors or confounders in both models simultaneously may not be always the most efficient, but is always valid (Tang *and others* (2024)).

From our updated analysis of Uminer study population with follow-up until 2012, after correcting

Table 4: Estimated Exposure Effects to Radon Progeny on Lung Cancer Mortality

	Age-adjusted model	
	$\frac{\lambda(t)}{\lambda_0(t)} = 1 + \beta_x x(t)$	$\frac{\lambda(t)}{\lambda_0(t)} = exp(\beta_x x(t))$
	ERR: β_x	Cox RR: $exp(\beta_x)$ (95% CI)
	$\Delta = 100$ WLM[2]	$\Delta = 100$ WLM
Uncorrected	1.61	1.46 [1.34, 1.59]
Additive ME[3]	5.28	6.11 [3.27, 11.42]
Multiplicative ME[4]	5.13	5.87 [2.94, 11.70]

	Multivariate model[1]	
	$\frac{\lambda(t)}{\lambda_0(t)} = exp(\boldsymbol{\beta}_{\boldsymbol{Z}}^{'}\mathbf{Z}(t))(1 + \beta_x x(t))$	$\frac{\lambda(t)}{\lambda_0(t)} = exp(\boldsymbol{\beta}_{\boldsymbol{Z}}^{'}\mathbf{Z}(t))exp(\beta_x x(t))$
	ERR: β_x	Cox RR: $exp(\beta_x)$ (95% CI)
	$\Delta = 100$ WLM	$\Delta = 100$ WLM
Uncorrected	0.75	1.35 [1.21, 1.50]
Additive ME[3]	2.83	4.46 [2.07, 9.63]
Multiplicative ME[4]	2.74	4.24 [1.81, 9.92]

**1. controlling for smoking (yes/no/missing), ethnic background (Hispanic/Native/White), company (5/6/12) and time-varying calendar year category (<1969/1970-1979/1980-1989/1990-1999/2000-2012) in both survival and measurement error models.

2. Unit for all exposure measurements is 100 WLM.

3. RRC correction for non-differential, additive linear, normal measurement error in 14 risk sets.

4. RRC correction for non-differential, multiplicative linear, log-normal measurement error in 14 risk sets.

for exposure measurement error, the estimated effect of every 100 WLM increments of cumulative total radon progeny exposure increased more than three-fold for both Cox-PH and ERR survival models. Differences between estimates assuming an additive and multiplication measurement error structure are not too significant. Compared with past literature (Table 5) reporting estimated ERR between radon progeny exposure and lung cancer mortality specifically adjusting for multiplicative measurement error, our estimates after RRC correction lie on the higher side.

6. Discussion

In this paper, we have extended the RRC method in three aspects: ERR survival model in addition to Cox survival model, multiplicative error model in addition to additive error model and main study/internal validation study design in addition to main study/external validation study design. We described a procedure to obtain error-corrected point and interval estimate of effect coefficients using the extended RRC method. We illustrated via various simulations that the extended RRC method is valid for all the scenarios under consideration and robust to different levels of error as well as certain types of model misspecification. Finally, we have applied this method to an updated Uminer cohort with followup until

Table 5: Select Literature Estimating ERR[1] (per 100 WLM) Between Radon Progeny and Lung Cancer Mortality Correcting for Multiplicative ME[2]

Paper	**Population**	**Adjustment**[3]	**Naive**	**Corrected**
Allodji *and others* (2012), Allodji *and others* (2012)	French Uranium Miners Cohort	Simulation extrapolation	0.7	1.8-1.9
		Substitution by expectation formula	0.776	1.098
		Regression calibration	0.776	0.971
		SIMEX	0.776	1.700
Hoffmann *and others* (2017)	Same as above, post 1955	Bayesian informative gamma	0.88	1.92
		Bayesian uniform	0.88	2.75
Heidenreich *and others* (2012)	Czech	Full likelihood maximization	2.4	2.4
	French		0.7	1.1
	German		0.45	0.46
Our results	New Mexico Uranium Miners Cohort	Extended risk set regression calibration	0.75	2.74

††1. Risk model is linear ERR model with additional Poisson assumption except for our results. Hoffman et al (2017) also included time since exposure and calendar year as effect modifier.

2. Exposure ME model is assumed to be non-differential, multiplicative linear, log-normal classical and/or Berkson error.

3. Adjustment uses prior or estimated knowledge on parameters in ME model, not external/internal validation study

2012 and produced an updated set of exposure effect measures which are on the higher scale compared to past literature but still within the reasonable range.

Due to limitations in the available data, we have defined 'gold-standard' annual exposure measurement to be individually-obtained exposure levels using the improved collection method after 1967. This metric is not perfect and may contain further bias from the true annual exposure an individual is exposed to. In addition, due to the development of exposure measurement technique, we only have pairs of surrogate and true annual exposure in validation study between calendar years 1967-1976. The estimated measurement error model from this validation set need to be extrapolated to cover calendar years prior to 1967 and after 1976. In our analysis, we inherently assumed that the measurement error models in each risk set defined by unique death age, are invariant to the passing of calendar years, which is not necessarily true.

In RRC correction procedure, when we identify risk sets in validation data, some risk sets, especially those defined by later event times, often contain too few data entries and error coefficient estimation based on those risk sets are unstable or even impossible. We opted to set a pre-determined number of data entries necessary to fit error model and combine the small risk sets until we have enough data entries in each risk set. When the total number of risk sets are relatively small, it is also possible to consider an optimal selection approach towards risk set combination strategy to achieve minimal estimation error in validation study. No matter the strategy, as long as some risk sets contain multiple event times, we

face the issue of correlation between repeated measurements in validation data. We employed GEE to fit error models taking correlation structure into consideration.

Throughout this work, we have treated all baseline covariates as possible risk factors in linear term and did not take into account possible effect modification, possible higher order terms as well as possible interaction terms. It could be of future interest to investigate further the performance of RRC correction method when the Cox-PH and ERR models extend beyond linear exposure effect terms and move into more complicated model structures.

Finally, our focus in this paper is entirely on a single cumulative total exposure variable, but the extended RRC framework could easily be generalized to accommodate a vector of exposure variables, both time-invariant and time-varying. Furthermore, in environmental and especially radiation epidemiology, there is evidence that exposure effect isn't immediate and isn't constant across time since exposure. There could be a lag period between exposure and first effect as well as dampened curve relationship between exposure and time-since-exposure. This is something that we intend to investigate further in future research.

7. Supplementary Materials

7.1 *Asymptotic Properties of Estimator $\hat{\boldsymbol{\beta}}_{RRC}$ Extended to ERR survival model and Multiplicative Error Model*

7.1.1 *Asymptotic Consistency*

Under the conditions for (3.7) to hold exactly, the asymptotic consistency and normality property of estimator $\hat{\boldsymbol{\beta}}_{RRC}$ for Cox-PH survival model and additive measurement error model has been proven by Liao *and others* (2011). We showcase here that similar proof could be expanded to ERR survival model and multiplicative measurement error model.

We first derive the partial log-likelihood function of the ERR survival model $\mathbf{l}(\boldsymbol{\beta})$.

$$
\begin{aligned}
L_j(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}}) &= \left\{ \frac{exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_j(T_j))(1+\beta_x\hat{x}_j(\hat{\boldsymbol{\alpha}},T_j))}{\sum_{i=1}^{n} Y_m(i,T_j)exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))} \right\}^{D_j} \\
\Rightarrow l_j(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}}) &= D_j \left[\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_j(T_j) + log(1+\beta_x\hat{x}_j(\hat{\boldsymbol{\alpha}},T_j)) - log(\sum_{i=1}^{n} Y_m(i,T_j)exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))) \right] \\
\Rightarrow \mathbf{l}(\boldsymbol{\beta}) &= \sum_{j=1}^{n_1} D_j \left[\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_j(T_j) + log(1+\beta_x\hat{x}_j(\hat{\boldsymbol{\alpha}},T_j)) - log(\sum_{i=1}^{n} Y_m(i,T_j)exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))) \right] \\
&= \sum_{i=1}^{n_1} \int_0^{T^*} \left[\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(t) + log(1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t)) - log(S^{(0)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)) \right] N_i(dt)
\end{aligned}
\tag{7.15}
$$

Then we derive the ERR score function $\mathbf{U}_{ERR}(\boldsymbol{\beta}) = \frac{\partial \mathbf{l}(\boldsymbol{\beta})}{\partial \boldsymbol{\beta}}$.

$$\frac{\partial l_j(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}_x} = D_j \left[\frac{\hat{x}_j(\hat{\boldsymbol{\alpha}},T_j)}{1+\beta_x \hat{x}_j(\hat{\boldsymbol{\alpha}},T_j)} - \frac{\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}} \mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)) \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)}}{\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}} \mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))} \right]$$

$$\frac{\partial l_j(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}_{\mathbf{Z}}} = D_j \left[\mathbf{Z}_j(T_j) - \frac{\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}} \mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)) \mathbf{Z}_i(t)}{\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}} \mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))} \right]$$

$$\Rightarrow \mathbf{U}_{ERR}(\boldsymbol{\beta}) = \sum_{i=1}^{n_1} \int_0^{T^*} \left\{ \begin{pmatrix} \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},t)} \\ \mathbf{Z}_i(t) \end{pmatrix} - \frac{\mathbf{S}^{(1)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)}{S^{(0)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)} \right\} N_i(dt) \tag{7.16}$$

For ERR survival model $\hat{\beta}_{x,RRC}$ will solve the following:

$$\sum_{i=1}^{n_1} \int_0^{T^*} \left\{ \begin{pmatrix} \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},t)} \\ \mathbf{Z}_i(t) \end{pmatrix} - \frac{\mathbf{S}^{(1)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)}{S^{(0)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t)} \right\} N_i(dt) = 0 \tag{7.17}$$

where

$$S^{(0)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t) = \frac{1}{n_1} \sum_{i=1}^{n_1} Y_m(i,t) exp(\boldsymbol{\beta}_{\boldsymbol{Z}} \mathbf{Z}(t))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},t)) \tag{7.18}$$

and

$$S^{(1)}_{ERR}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t) = \frac{1}{n_1} \sum_{i=1}^{n_1} Y_m(i,t) \begin{pmatrix} \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},t)} \\ \mathbf{Z}_i(t) \end{pmatrix} exp(\boldsymbol{\beta}_{\boldsymbol{Z}} \mathbf{Z}(t))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},t)) \tag{7.19}$$

We further write out the calculations $\frac{\partial \mathbf{U}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}}$.

$$\frac{\partial l_j^2(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}_x^2} = D_j \left[\frac{-\hat{x}_j^2(\hat{\boldsymbol{\alpha}},T_j)}{[1+\beta_x \hat{x}_j(\hat{\boldsymbol{\alpha}},T_j)]^2} - \frac{-[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)) \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)}]^2}{[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))]^2} \right]$$

$$\frac{\partial l_j^2(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}_x \partial \boldsymbol{\beta}_{\mathbf{Z}}} = \frac{\partial l_j^2(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}_{\mathbf{Z}} \partial \boldsymbol{\beta}_x} = -D_j$$

$$\left\{ \frac{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)) \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)} \mathbf{Z}_{\mathbf{i}}(T_j)\right] \left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\right]}{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\right]^2} \right\}$$

$$- \left\{ \frac{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)) \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)}\right] \left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\mathbf{Z}_{\mathbf{i}}(T_j)\right]}{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\right]^2} \right\}$$

$$\frac{\partial l_j^2(\boldsymbol{\beta};\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}_{\mathbf{Z}}^2} = -D_j \left\{ \frac{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\mathbf{Z}_i^2(t)\right] \left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\right]}{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\right]^2} \right\}$$

$$- \left\{ \frac{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\mathbf{Z}_i(t)\right]^2}{\left[\sum_{i=1}^{n} Y_m(i,T_j) exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}_i(T_j))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))\right]^2} \right\} \tag{7.20}$$

Now we define

$$S_{ERR}^{(2)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},t) = \frac{1}{n_1}\sum_{i=1}^{n_1} Y_m(i,t) \begin{pmatrix} \frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},t)} \\ \mathbf{Z}_i(t) \end{pmatrix}^{\otimes 2} exp(\boldsymbol{\beta}_{\boldsymbol{Z}}\mathbf{Z}(t))(1+\beta_x \hat{x}_i(\hat{\boldsymbol{\alpha}},t)) \tag{7.21}$$

and we are able to write all score function and their derivatives in similar combinations of $S_{ERR}^{(0)}, S_{ERR}^{(1)}$ and $S_{ERR}^{(2)}$, see Liao *and others* (2011) for comparisons with Cox-PH format.

Following the same logic as Liao *and others* (2011) supplementary materials, we are able to conclude that under certain regularity conditions, $\frac{\mathbf{L}(\boldsymbol{\beta})}{n_1} \xrightarrow{P} \mathbf{H}(\boldsymbol{\beta})$ where

$$\mathbf{H}(\boldsymbol{\beta}) = \int_0^{T^*} [\boldsymbol{\beta} s^{(1)}(t) - log(s^{(0)}(\boldsymbol{\beta},t)) s^{(0)}(t)] dt \tag{7.22}$$

and we have $\hat{\boldsymbol{\beta}}_{RRC} \xrightarrow{P} \boldsymbol{\beta}^*$ which maximizes $\mathbf{H}(\boldsymbol{\beta})$.

7.1.2 *Asymptotic Normality*

Since the measurement error model coefficients $\hat{\boldsymbol{\alpha}}$ are obtained by fitting regression models in the validation study data, we need to combine variability of the estimated measurement error model coefficients with the variability of the estimated survival model effect coefficients. Denote the true value of measurement error model coefficients as $\boldsymbol{\alpha}^*$. We use Taylor's theorem to expand the estimating score equation

around true value of measurement error and survival coefficients $(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*)$.

$$\begin{aligned} 0 = \mathbf{U}_{ERR}(\hat{\boldsymbol{\beta}}_{RRC}, \hat{\boldsymbol{\alpha}}) &\approx \mathbf{U}_{ERR}(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*) + \frac{\partial \mathbf{U}_{ERR}(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*)}{\partial \boldsymbol{\beta}}(\hat{\boldsymbol{\beta}}_{RRC} - \boldsymbol{\beta}^*) + \frac{\partial \mathbf{U}_{ERR}(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) \\ \Rightarrow n_1^{\frac{1}{2}}(\hat{\boldsymbol{\beta}}_{RRC} - \boldsymbol{\beta}^*) &\approx \left[-n_1^{-1}\frac{\partial \mathbf{U}_{ERR}(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*)}{\partial \boldsymbol{\beta}}\right]^{-1} \cdot n_1^{\frac{1}{2}}\left[\mathbf{U}_{ERR}(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*) + \frac{\partial \mathbf{U}_{ERR}(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*)}{\partial \boldsymbol{\beta}}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*)\right] \end{aligned} \tag{7.23}$$

Firstly, define

$$\hat{\mathbf{I}}(\boldsymbol{\beta}^*) = -n_1^{-1}\frac{\partial \mathbf{U}_{ERR}(\boldsymbol{\beta}^*, \boldsymbol{\alpha}^*)}{\partial \boldsymbol{\beta}} \tag{7.24}$$

then it could be verified by Andersen and Gill (1982) that $\hat{\mathbf{I}}(\boldsymbol{\beta}^*) \xrightarrow{P} \mathbf{I}(\boldsymbol{\beta}^*)$ and then $\mathbf{I}(\boldsymbol{\beta}^*)$ could be estimated with $\hat{\mathbf{I}}_{\boldsymbol{\beta}}$ in equation.

Secondly, following similar proof to Lin and Wei (1989) and multivariate central limit theorem, we have $n_1^{-\frac{1}{2}}\mathbf{U}(\boldsymbol{\beta}^*) \xrightarrow{D} N(\mathbf{0}, \mathbb{E}[\mathbf{G}_i(\boldsymbol{\beta}^*)^{\otimes 2}])$ where

$$\begin{aligned} \mathbf{G}_i(\boldsymbol{\beta}^*) = &\int_0^{T^*}\left\{\begin{pmatrix}\frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t)} \\ \mathbf{Z}_i(t)\end{pmatrix} - \frac{\mathbf{S}^{(1)}_{ERR}(\boldsymbol{\beta}^*, \hat{\boldsymbol{\alpha}}, t)}{S^{(0)}_{ERR}(\boldsymbol{\beta}^*, \hat{\boldsymbol{\alpha}}, t)}\right\} N_i(dt) \\ &- \int_0^{T^*}\frac{Y_m(i,t)exp(\boldsymbol{\beta}^*_{\mathbf{Z}}\mathbf{Z}(t))(1+\beta_x^*\hat{x}_i(\hat{\boldsymbol{\alpha}},t))}{S^{(0)}_{ERR}(\boldsymbol{\beta}^*, \hat{\boldsymbol{\alpha}}, T_j)} \cdot \left\{\begin{pmatrix}\frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},t)}{1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},t)} \\ \mathbf{Z}_i(t)\end{pmatrix} - \frac{\mathbf{S}^{(1)}_{ERR}(\boldsymbol{\beta}^*, \hat{\boldsymbol{\alpha}}, t)}{S^{(0)}_{ERR}(\boldsymbol{\beta}^*, \hat{\boldsymbol{\alpha}}, t)}\right\}\tilde{F}(dt) \end{aligned} \tag{7.25}$$

with $\tilde{F}(t) = \mathbb{E}[\sum_{i=1}^{n_1} N_i(t)/n_1]$ and then $\mathbb{E}[\mathbf{G}_i(\boldsymbol{\beta}^*)^{\otimes 2}]$ could be estimated with $\mathbf{H}_{\boldsymbol{\beta}}$ in equation.

Finally using Taylor's theorem to expand the estimating score equation for measurement error model around true value of measurement error coefficients $\boldsymbol{\alpha}^*$, we obtain:

$$\begin{aligned} \mathbf{0} = \mathbf{U}_{\boldsymbol{\alpha}}(\hat{\boldsymbol{\alpha}}) &\approx \mathbf{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}^*) + \frac{\partial \mathbf{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) \\ \Rightarrow n_2^{\frac{1}{2}}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) &\approx -\left[\frac{1}{n_2}\frac{\partial \mathbf{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}^*)}{\partial \boldsymbol{\alpha}}\right] n_2^{-\frac{1}{2}}\mathbf{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}^*) \end{aligned} \tag{7.26}$$

following similar proof as before, we have $n_2^{\frac{1}{2}}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) \xrightarrow{D} N(\mathbf{0}, \mathbb{E}[\mathbf{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}^*)^{\otimes 2}])$ and then $\mathbb{E}[\mathbf{U}_{\boldsymbol{\alpha}}(\boldsymbol{\alpha}^*)^{\otimes 2}]$ could be estimated with $\hat{\mathbf{V}}_{\hat{\boldsymbol{\alpha}}}$ in equation.

Combining all intermediate steps, we then have $n_1^{\frac{1}{2}}(\hat{\boldsymbol{\beta}}_{RRC} - \boldsymbol{\beta}^*) \xrightarrow{D} N(0, V(\boldsymbol{\beta}^*))$, where $V(\boldsymbol{\beta}^*)$ could be consistently estimated with the formula in equation.

7.1.3 *Variance Formula of $\hat{\boldsymbol{\beta}}_{RRC}$*

In the main study, $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_{\mathbf{Z}}, \hat{\beta}_x)$ solves the score equation which can be rewritten as $\sum_{i=1}^{n_1}\mathbf{U}_{\boldsymbol{\beta}i}(\boldsymbol{\beta}|\hat{\boldsymbol{\alpha}}) = 0$, where for Cox-PH model

$$\mathbf{U}_{Cox,\boldsymbol{\beta}i}(\boldsymbol{\beta}|\hat{\boldsymbol{\alpha}}) = D_i\left\{\begin{pmatrix}\hat{x}_i(\hat{\boldsymbol{\alpha}}, T_i) \\ \mathbf{Z}_i(T_i)\end{pmatrix} - \frac{\mathbf{S}^{(1)}_{Cox}(\boldsymbol{\beta}, \hat{\boldsymbol{\alpha}}, T_i)}{S^{(0)}_{Cox}(\boldsymbol{\beta}, \hat{\boldsymbol{\alpha}}, T_i)}\right\} \tag{7.27}$$

and for ERR model

$$\mathbf{U}_{ERR,\boldsymbol{\beta}i}(\boldsymbol{\beta}|\hat{\boldsymbol{\alpha}}) = D_i\left\{\begin{pmatrix}\frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},T_i)}{1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},T_i)} \\ \mathbf{Z}_i(T_i)\end{pmatrix} - \frac{\mathbf{S}^{(1)}_{ERR}(\boldsymbol{\beta}, \hat{\boldsymbol{\alpha}}, T_i)}{S^{(0)}_{ERR}(\boldsymbol{\beta}, \hat{\boldsymbol{\alpha}}, T_i)}\right\} \tag{7.28}$$

In the validation study, $\hat{\boldsymbol{\alpha}} = (\hat{\boldsymbol{\alpha}}(t_{k_1}), \ldots, \hat{\boldsymbol{\alpha}}(t_{k_R}))$ solves the score equation which can be rewritten as $\sum_{i=1}^{n_2} \mathbf{U}_{\boldsymbol{\alpha} i,r}(\boldsymbol{\alpha}) = 0$ and further define $\mathbf{U}_{\boldsymbol{\alpha} i} = (\mathbf{U}_{\boldsymbol{\alpha} i,1}, \ldots, \mathbf{U}_{\boldsymbol{\alpha} i,R})$, where $\mathbf{U}_{\boldsymbol{\alpha} i,r}(\boldsymbol{\alpha}) = (\mathbf{U}_{\boldsymbol{\alpha}_0 i,r}(\boldsymbol{\alpha}), \mathbf{U}_{\boldsymbol{\alpha}_1 i,r}(\boldsymbol{\alpha}), \mathbf{U}_{\boldsymbol{\alpha}_2 i,r}(\boldsymbol{\alpha}))$ for additive measurement error model

$$\begin{aligned}
\mathbf{U}_{\boldsymbol{\alpha}_0 i,r}(\boldsymbol{\alpha}) &= Y_v(i,t_{k_r})\left[c_i(t_{k_r}) - \alpha_0(t_{k_r}) - \alpha_1(t_{k_r})C_i(t_{k_t}) - \boldsymbol{\alpha}_2(t_{k_r})\mathbf{Z}_i(t_{k_t})\right] \\
\mathbf{U}_{\boldsymbol{\alpha}_1 i,r}(\boldsymbol{\alpha}) &= Y_v(i,t_{k_r})C_i(t_{k_r})\left[c_i(t_{k_r}) - \alpha_0(t_{k_r}) - \alpha_1(t_{k_r})C_i(t_{k_t}) - \boldsymbol{\alpha}_2(t_{k_r})\mathbf{Z}_i(t_{k_t})\right] \\
\mathbf{U}_{\boldsymbol{\alpha}_2 i,r}(\boldsymbol{\alpha}) &= Y_v(i,t_{k_r})\mathbf{Z}_i(t_{k_r})\left[c_i(t_{k_r}) - \alpha_0(t_{k_r}) - \alpha_1(t_{k_r})C_i(t_{k_t}) - \boldsymbol{\alpha}_2(t_{k_r})\mathbf{Z}_i(t_{k_t})\right]
\end{aligned} \tag{7.29}$$

and for multiplicative measurement error model

$$\begin{aligned}
\mathbf{U}_{\boldsymbol{\alpha}_0 i,r}(\boldsymbol{\alpha}) &= Y_v(i,t_{k_r})\left[log(c_i(t_{k_r})) - \alpha_0(t_{k_r}) - \alpha_1(t_{k_r})log(C_i(t_{k_t})) - \boldsymbol{\alpha}_2(t_{k_r})log(\mathbf{Z}_i(t_{k_t}))\right] \\
\mathbf{U}_{\boldsymbol{\alpha}_1 i,r}(\boldsymbol{\alpha}) &= Y_v(i,t_{k_r})log(C_i(t_{k_r}))\left[log(c_i(t_{k_r})) - \alpha_0(t_{k_r}) - \alpha_1(t_{k_r})log(C_i(t_{k_t})) - \boldsymbol{\alpha}_2(t_{k_r})log(\mathbf{Z}_i(t_{k_t}))\right] \\
\mathbf{U}_{\boldsymbol{\alpha}_2 i,r}(\boldsymbol{\alpha}) &= Y_v(i,t_{k_r})log(\mathbf{Z}_i(t_{k_r}))\left[log(c_i(t_{k_r})) - \alpha_0(t_{k_r}) - \alpha_1(t_{k_r})log(C_i(t_{k_t})) - \boldsymbol{\alpha}_2(t_{k_r})log(\mathbf{Z}_i(t_{k_t}))\right]
\end{aligned} \tag{7.30}$$

Regardless of which form of measurement error model we choose or which modeling method we use to fit such model, we need to obtain an approximate estimate of $Cov(\hat{\boldsymbol{\alpha}})$, which following Taylor's theorem and ensuing asymptotic normality property, can be estimated using the sandwich variance formula $\hat{\mathbf{V}}_{\hat{\boldsymbol{\alpha}}}$ given below.

Following derivations in Liao *and others* (2011) supplementary materials, we obtain the following formula for the variance of RRC estimators:

$$\hat{Var}(\hat{\boldsymbol{\beta}}_{x,RRC}) = \frac{1}{n_1}\hat{I}_{\boldsymbol{\beta}}^{-1}\hat{H_{\boldsymbol{\beta},\boldsymbol{\alpha}}}\hat{I}_{\boldsymbol{\beta}}^{-1} \tag{7.31}$$

where

$$\hat{I}_{\boldsymbol{\beta}}^{-1} = \left[\frac{1}{n_1}\sum_{i=1}^{n_1}\frac{\partial \mathbf{U}_{\boldsymbol{\beta} i}(\boldsymbol{\beta}|\hat{\boldsymbol{\alpha}})}{\partial \boldsymbol{\beta}}\right]^{-1}|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}_{RRC}} \tag{7.32}$$

$$\hat{H_{\boldsymbol{\beta},\boldsymbol{\alpha}}} = \hat{H}_{\hat{\boldsymbol{\beta}}} + \frac{1}{n_1 n_2}\hat{\boldsymbol{U}}^*(\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\alpha}})\hat{\mathbf{V}}_{\hat{\boldsymbol{\alpha}}}\hat{\boldsymbol{U}}^*(\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\alpha}})^T \tag{7.33}$$

in which

$$\hat{\boldsymbol{U}}^*(\hat{\boldsymbol{\beta}},\hat{\boldsymbol{\alpha}}) = \sum_{i=1}^{n_1}\frac{\partial \mathbf{U}_{\boldsymbol{\beta}_i}(\boldsymbol{\beta}|\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}}|_{(\boldsymbol{\beta},\boldsymbol{\alpha})=(\hat{\boldsymbol{\beta}}_{RRC},\hat{\alpha})} \tag{7.34}$$

$$\hat{\mathbf{V}}_{\hat{\boldsymbol{\alpha}}} = \left[\frac{1}{n_2}\sum_{i=1}^{n_2}\frac{\partial \mathbf{U}_{\boldsymbol{\alpha} i}(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}}\right]^{-1}\left[\frac{1}{n_2}\sum_{i=1}^{n_2}\mathbf{U}_{\boldsymbol{\alpha} i}(\boldsymbol{\alpha})\otimes\mathbf{U}_{\boldsymbol{\alpha} i}^T(\boldsymbol{\alpha})\right]\left[\frac{1}{n_2}\sum_{i=1}^{n_2}\frac{\partial \mathbf{U}_{\boldsymbol{\alpha} i}(\boldsymbol{\alpha})}{\partial \boldsymbol{\alpha}}\right]^{-1}|_{\boldsymbol{\alpha}=\hat{\boldsymbol{\alpha}}} \tag{7.35}$$

$$\hat{H}_{\hat{\boldsymbol{\beta}}} = \frac{1}{n_1}\sum_{i=1}^{n_1}\tilde{\mathbf{U}}_{\boldsymbol{\beta}_i}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}})\tilde{\mathbf{U}}_{\boldsymbol{\beta}_i}^T(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}})|_{\boldsymbol{\beta}=\hat{\boldsymbol{\beta}}_{RRC}} \tag{7.36}$$

with

$$\tilde{\mathbf{U}}_{\boldsymbol{\beta}_i}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}}) = \mathbf{U}_{\boldsymbol{\beta}_i}(\boldsymbol{\beta}|\hat{\boldsymbol{\alpha}}) - \sum_{j=1}^{n_1}\frac{D_j Y_m(i,T_j)exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}(T_j))exp(\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))}{n_1 S_{Cox}^{(0)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_j)}\cdot\left\{\begin{pmatrix}\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j)\\ \mathbf{Z}_i(T_j)\end{pmatrix} - \frac{\mathbf{S}_{Cox}^{(1)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_j)}{S_{Cox}^{(0)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_j)}\right\} \tag{7.37}$$

for Cox-PH model and

$$\tilde{\mathbf{U}}_{\boldsymbol{\beta}_i}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}}) = \mathbf{U}_{\boldsymbol{\beta}_i}(\boldsymbol{\beta}|\hat{\boldsymbol{\alpha}}) - \sum_{j=1}^{n_1}\frac{D_j Y_m(i,T_j)exp(\boldsymbol{\beta}_{\mathbf{Z}}\mathbf{Z}(T_j))(1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},T_j))}{n_1 S_{ERR}^{(0)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_j)}\cdot\left\{\begin{pmatrix}\frac{\hat{x}_i(\hat{\boldsymbol{\alpha}},T_i)}{1+\beta_x\hat{x}_i(\hat{\boldsymbol{\alpha}},T_i)}\\ \mathbf{Z}_i(T_i)\end{pmatrix} - \frac{\mathbf{S}_{ERR}^{(1)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_i)}{S_{ERR}^{(0)}(\boldsymbol{\beta},\hat{\boldsymbol{\alpha}},T_i)}\right\} \tag{7.38}$$

for ERR model.

7.2 *Concept of Comparable Cox-PH and ERR Effect Coefficient*

We have written the Cox-PH effect model and ERR effect models respectively as:

$$\lambda_C(t) = \lambda_C(t; \{c(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}) = \lambda_{0,C}(t) exp(\boldsymbol{\beta_{Z,C}} Z(t)) exp(\beta_{x,C} x(t)) \triangleq \lambda_{0,C}^{'}(t) exp(\beta_{x,C} x(t))$$

and

$$\lambda_E(t) = \lambda_E(t; \{c(v), v \leqslant t\}, \{\boldsymbol{Z}(v), v \leqslant t\}) = \lambda_{0,E}(t) exp(\boldsymbol{\beta_{Z,E}} Z(t))(1 + \beta_{x,E} x(t)) \triangleq \lambda_{0,E}^{'}(t)(1 + \beta_{x,E} x(t))$$

We introduce the concept of a pair of 'comparable' Cox-PH and ERR effect coefficients which we employ in our simulations. We aim for an approximate comparability between hazard functions, i.e.

$$\begin{aligned} &\lambda_C(t) = \lambda_E(t) \\ \Rightarrow &\lambda_{0,C}^{'}(t) exp(\beta_{x,C} x(t)) = \lambda_{0,E}^{'}(t)(1 + \beta_{x,E} x(t)) \\ \Rightarrow &\beta_{x,E} = \frac{e^{\beta_{x,C} x(t)} - 1}{x(t)} \end{aligned}$$

under the assumption that baseline hazard rate under both Cox-PH and ERR models are equivalent, which gives us $\lambda_C(t)|_{x(t)=0} = \lambda_{0,C}^{'}(t, \boldsymbol{Z}(t)) = \lambda_{0,E}^{'}(t, \boldsymbol{Z}(t)) = \lambda_E(t)|_{x(t)=0}$.

This transition formula between a pair of 'comparable' Cox-PH and ERR models depends on the value of cumulative total exposure $x(t)$ at a given time t, therefore exact solution is varying across the distribution of cumulative total exposure. For purposes of simulation, we plug in the median of generated $x(t)$ values and obtain a pair of approximately comparable Cox-PH and ERR coefficients.

Illustrating this concept through a real simulation scenario. In a generated simulation data set with median$(x(t)) = 0.204$, the comparable ERR coefficient to a Cox-PH coefficient of $\beta_{x,C} = 1$ is $\beta_{x,E} \approx \frac{e^{\beta_{x,C} median(x(t))} - 1}{median(x(t))} \approx 1.109$.

7.3 *Additional Simulation 1 Results*

Results presented in Section 4.1 Table 1 is when true underlying exposure effect is relatively small $\beta_{Cox} \approx 0.405$, satisfying the loose 'small effect' assumption needed to successfully apply RRC correction method to Cox-PH survival model.

We further look at finite-sample simulation results for when the loose 'small effect' assumption is broken, namely when $\beta_{Cox} = log(5) \approx 1.609$ and comparable $\beta_{ERR} \approx 1.905$. We adjust (ν, θ) parameters in baseline hazard function to main a similar range of cumulative incidence rate between $5 - 10\%$. Simulation summary results are presented in Table 6.

We observe similar trends with respect to the validity of RRC correction method and how that changes as correlation structure between repeated measurements get more and more complicated. Additionally, comparing results in Table 6 to Table 1, we note that the performance of RRC correction declines noticeably for Cox-PH survival models, when the 'small effect' assumption no longer holds, while declines slightly for ERR survival models.

Table 6: Summary Results of Simulation Study of Naive VS RRC-Corrected Effect Estimates of Cumulative Total Exposure

Generation of C	**Relative bias (%)**		**Estimated SE $\hat{se}(\hat{\beta})$**		**SE/SD**		**95% CI coverage (%)**	
	Naive	RRC	Naive	RRC	Naive	RRC	Naive	RRC
Cox+Additive	Cox true $\beta = log(5) \approx 1.609$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-50.42	-6.44	0.062	0.098	0.997	0.993	0.00	91.25
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-57.22	-7.28	0.065	0.112	0.996	0.992	0.00	91.10
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-57.58	-7.93	0.064	0.112	0.996	0.992	0.00	91.05
Resample from data	-59.01	-9.88	0.069	0.126	0.994	0.991	0.00	90.70
Cox+Multiplicative	Cox true $\beta = log(5) \approx 1.609$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-51.11	-6.62	0.068	0.100	0.997	0.994	0.00	91.15
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-57.64	-7.49	0.070	0.115	0.996	0.993	0.00	90.90
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-57.79	-8.17	0.071	0.119	0.995	0.992	0.00	90.80
Resample from data	-59.08	-9.96	0.074	0.126	0.993	0.991	0.00	90.15
Cox and ERR true β are chosen to produce approximately comparable risk model								
ERR+Additive	ERR true $\beta \approx 1.905$, additive independent measurement error							
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-52.05	-4.91	0.088	0.115	0.998	0.995	0.00	92.75
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-57.96	-6.36	0.094	0.117	0.998	0.995	0.00	92.55
MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-58.10	-7.18	0.097	0.118	0.998	0.994	0.00	92.40
Resample from data	-61.12	-9.63	0.121	0.126	0.996	0.992	0.00	91.85
ERR+Multiplicative	ERR true $\beta \approx 1.905$, multiplicative independent measurement error							
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{ind}}$)	-50.12	-5.06	0.090	0.116	0.997	0.992	0.00	92.35
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{AR(1)}}$)	-55.56	-7.42	0.101	0.122	0.997	0.991	0.00	92.05
Log-MVN($\boldsymbol{\mu_C}, \boldsymbol{\Sigma}_{\text{DEX}}$)	-56.03	-8.15	0.099	0.119	0.996	0.991	0.00	92.00
Resample from data	-58.87	-10.07	0.130	0.134	0.995	0.990	0.00	91.45

[‡‡]Cox-PH survival model $\lambda(t) = \lambda_0(t)exp(\beta x(t))$ with true $\beta_{Cox} = log(5) \approx 1.609$ and comparable ERR survival model $\lambda(t) = \lambda_0(t)(1 + \beta x(t))$ with true $\beta_{ERR} \approx 1.905$
Main study sample size n_1=2400, Validation study sample size n_2=800, Study duration T=55, Simulation repetition number B=2000, CIR between $5 - 10\%$

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